

Exploratory Data Analysis - Singapore Total Fertility Rate & Total Live Births

Hunter Holt

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Introduction

This exploratory data analysis (EDA) investigates the temporal features of two significant population indicators for Singapore:

- **Total Fertility Rate (TFR):** the average number of children a woman would have over her lifetime at current age-specific fertility rates;
- **Total Live Births (TLB):** the total number of live births recognised per year.

The dates for the SingaporeCensus.csv reference dataset span from 1960 to 2024, sourced from the Singapore Department of Statistics. The objective is to recognise and analyse long-run trends, assess stationarity, and identify candidate time series models for a proposed training period 1960 - 2012 and a proposed forecasting performance assessed over the remaining 2013 - 2024 period.

Data Loading & Reshaping

The given data is in a CSV file, which is optimal for computation. However, the data is in a ‘wide format’, where its temporal variables (i.e. years) are given as columns instead of years. We transpose the data and extract the two data columns we are interested in.

Below is the transposition framework of the data.

Wide format (raw CSV)

DataSeries		1960		1961		...		2024
TFR row								
TLB row								

pivot_longer()

+

filter()

+

arrange()

Long format (processed)

Series		Year		Value
TFR		1960		5.76
TFR		1961		5.45
TLB		1960		61775
TLB		1961		62157

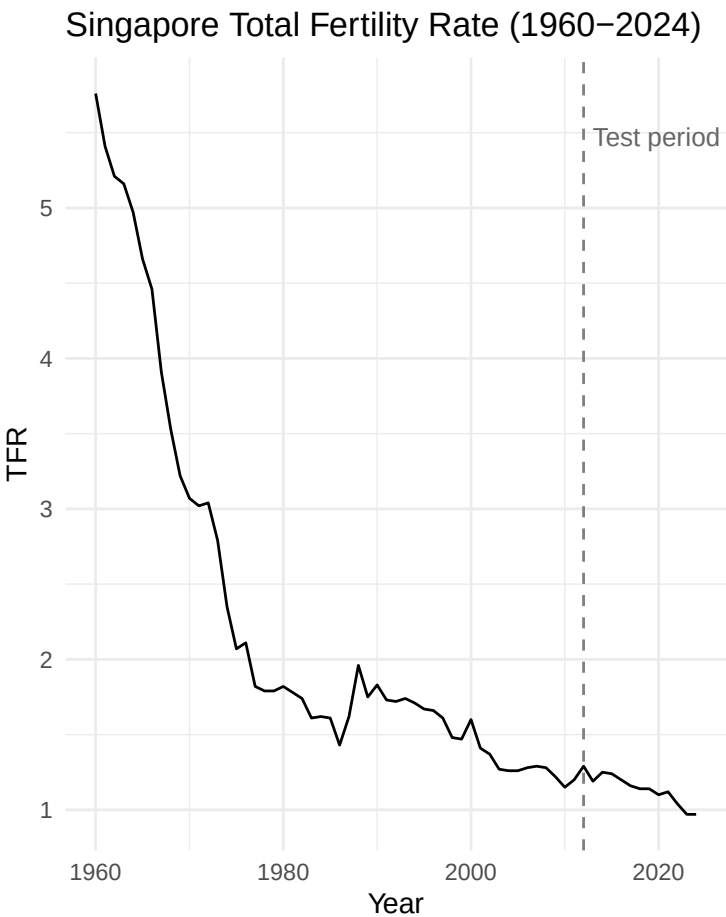
Data Quality & Summary

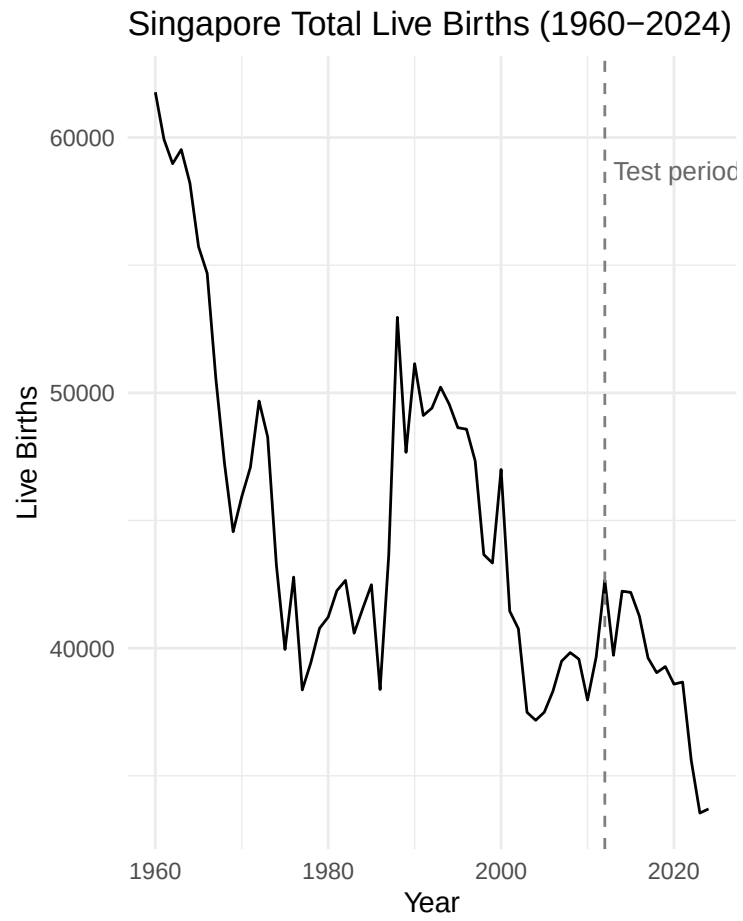
Table 1: Summary statistics for TFR and TLB (1960-2024)

Statistic	TFR	TLB
Min	0.97	33541.00
Q1	1.26	39570.00
Median	1.62	42484.00
Mean	2.06	44359.63
Q3	2.07	48577.00
Max	5.76	61775.00
NAs	0.00	0.00

Time Series Plots

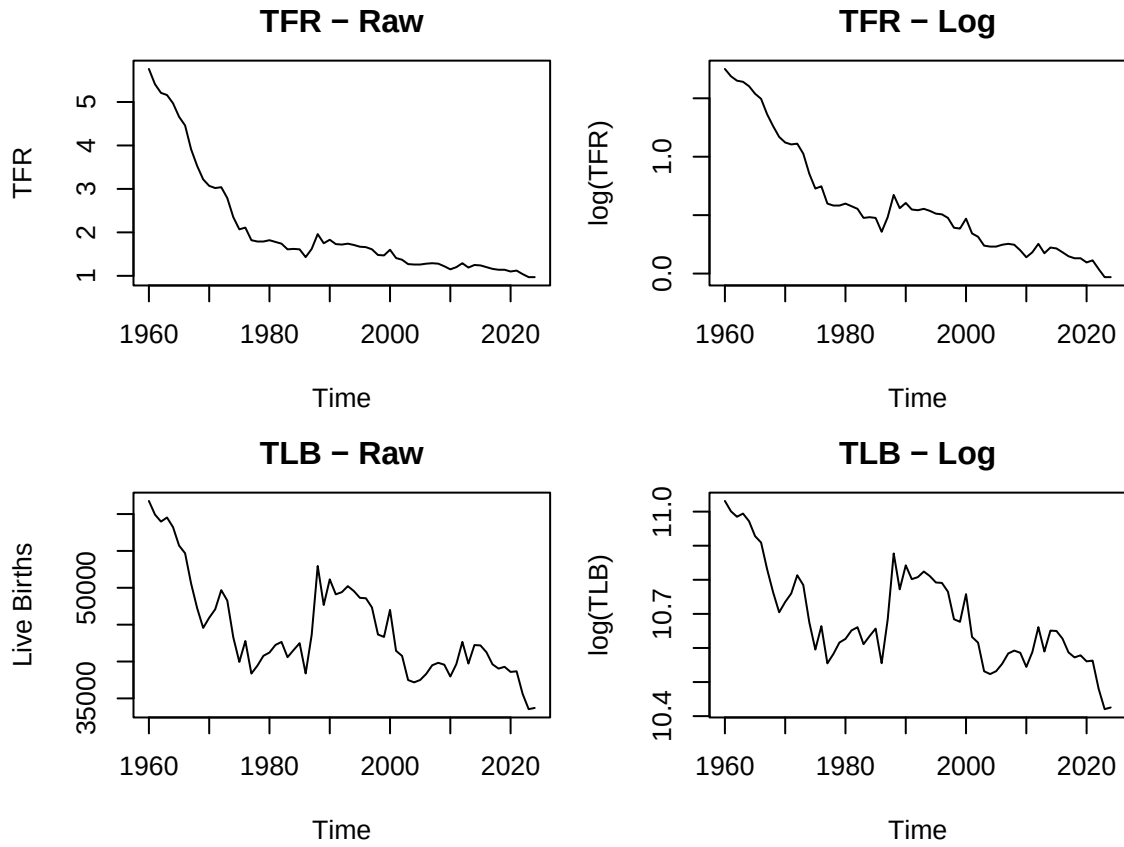
Below, the dashed vertical line marks the train-test split at 2012.





Temporal Feature Evaluation

Log Transformation Assessment



Stationarity Testing

Raw Series

Table 2: Stationarity tests - raw training series (1960-2012)

Series	ADF p-value	KPSS p-value
TFR	0.120	0.0100
TLB	0.356	0.0205

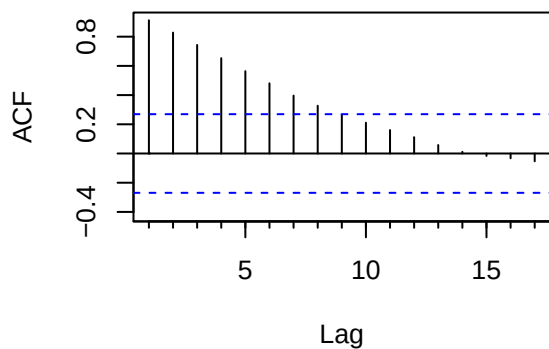
First Difference Series

Table 3: Stationarity tests - first-differenced training series

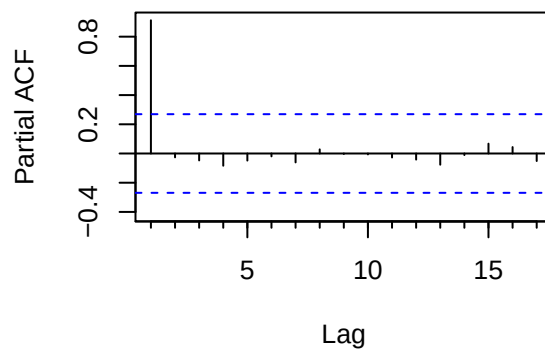
Series	ADF p-value	KPSS p-value
Δ TFR	0.0830	0.01
Δ TLB	0.0395	0.10

ACF & PACF Analysis

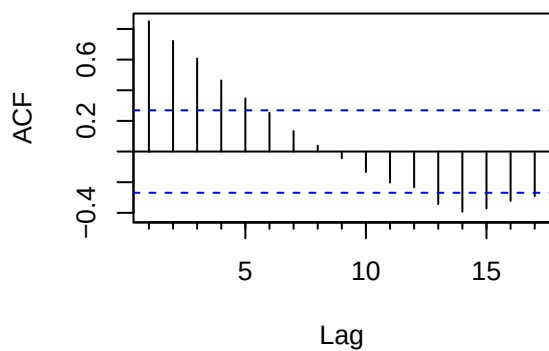
TFR - ACF



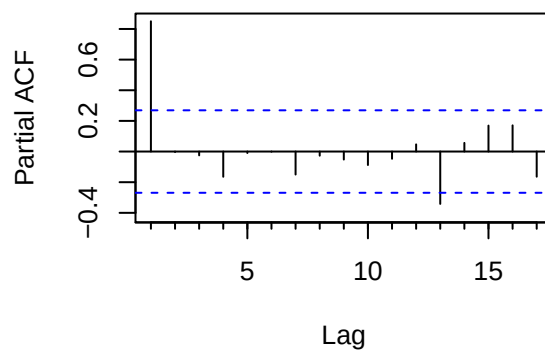
TFR - PACF

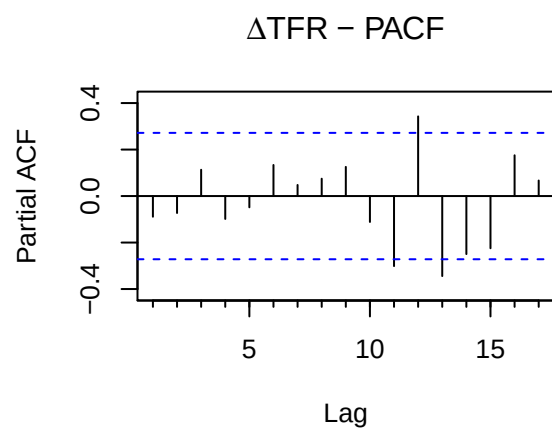
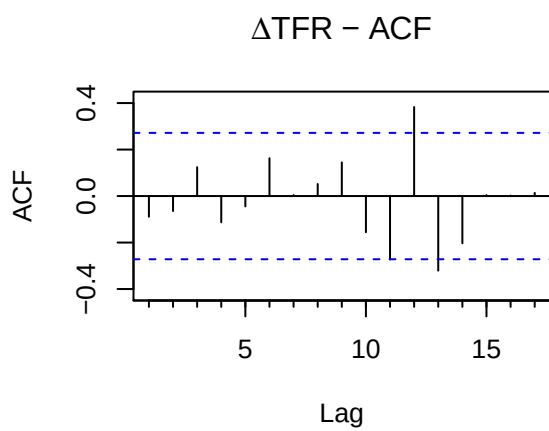
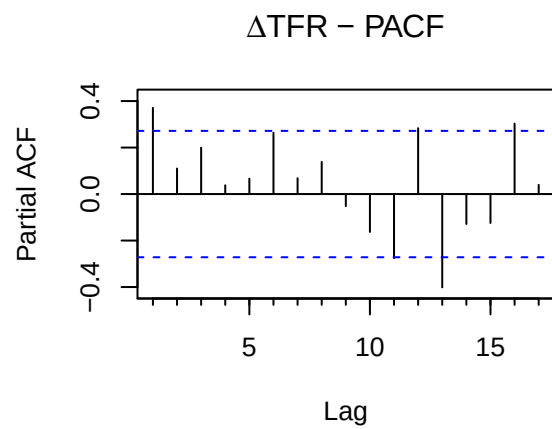
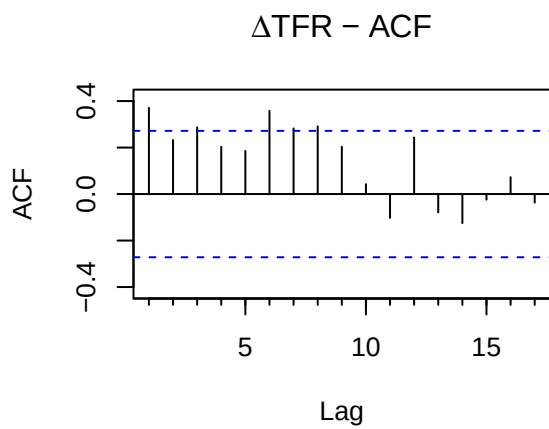


TLB - ACF

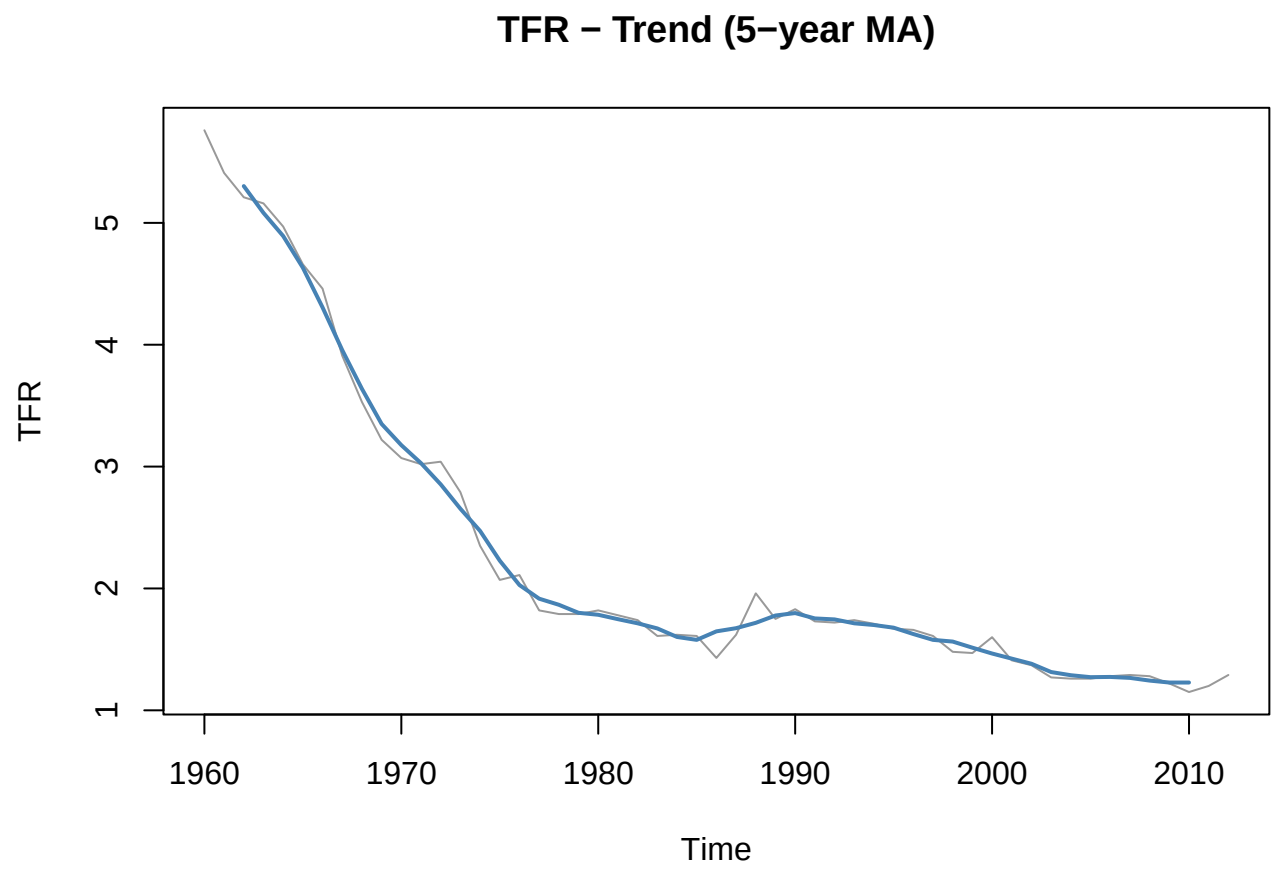


TLB - PACF





Decomposition



TLB – Trend (5-year MA)

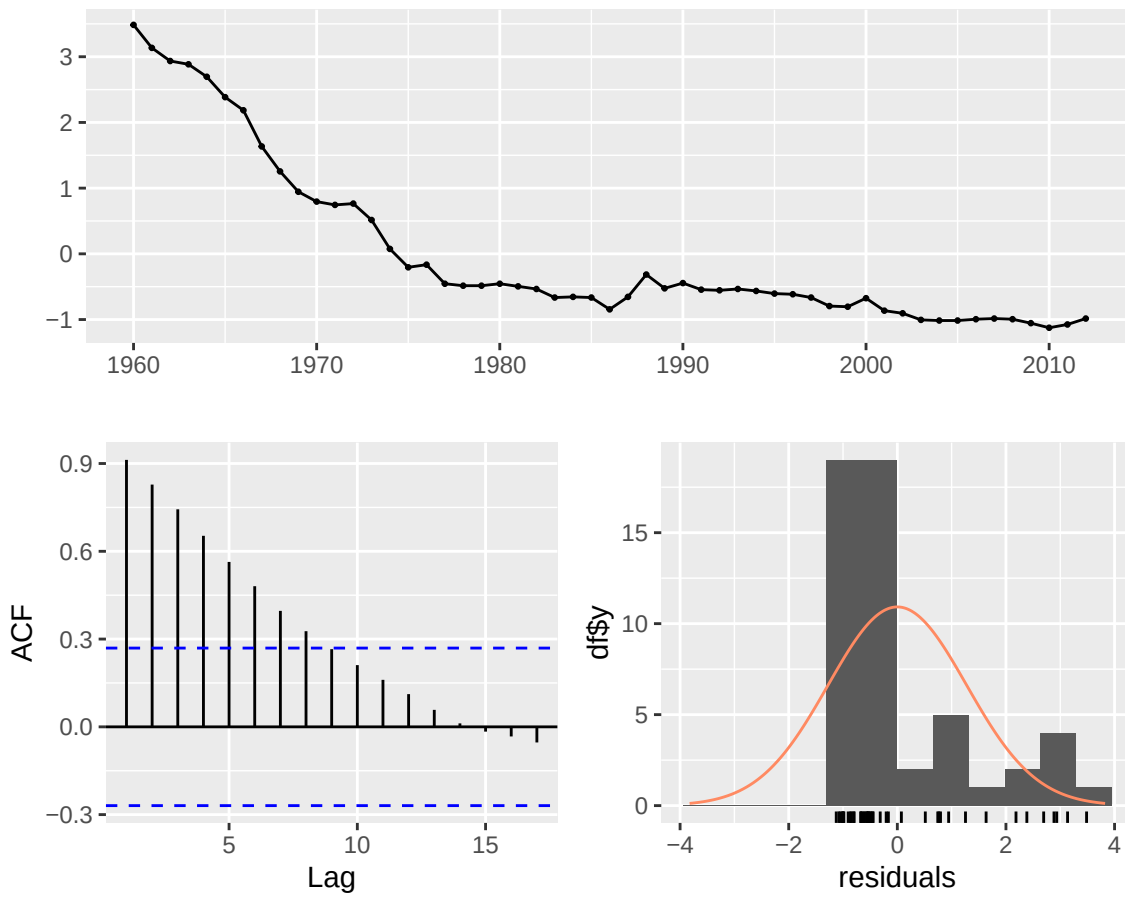


Candidate Model Fitting

TFR - Candidate Model

```
## Series: tfr_train_ts
## ARIMA(0,0,0) with non-zero mean
##
## Coefficients:
##      mean
##      2.2751
## s.e.  0.1735
##
## sigma^2 = 1.626: log likelihood = -87.58
## AIC=179.17  AICc=179.41  BIC=183.11
##
## Training set error measures:
##              ME      RMSE      MAE      MPE      MAPE      MASE
## Training set 4.127809e-13 1.263122 0.9974938 -23.15142 45.38353 7.943289
##              ACF1
## Training set 0.9124029
```

Residuals from ARIMA(0,0,0) with non-zero mean

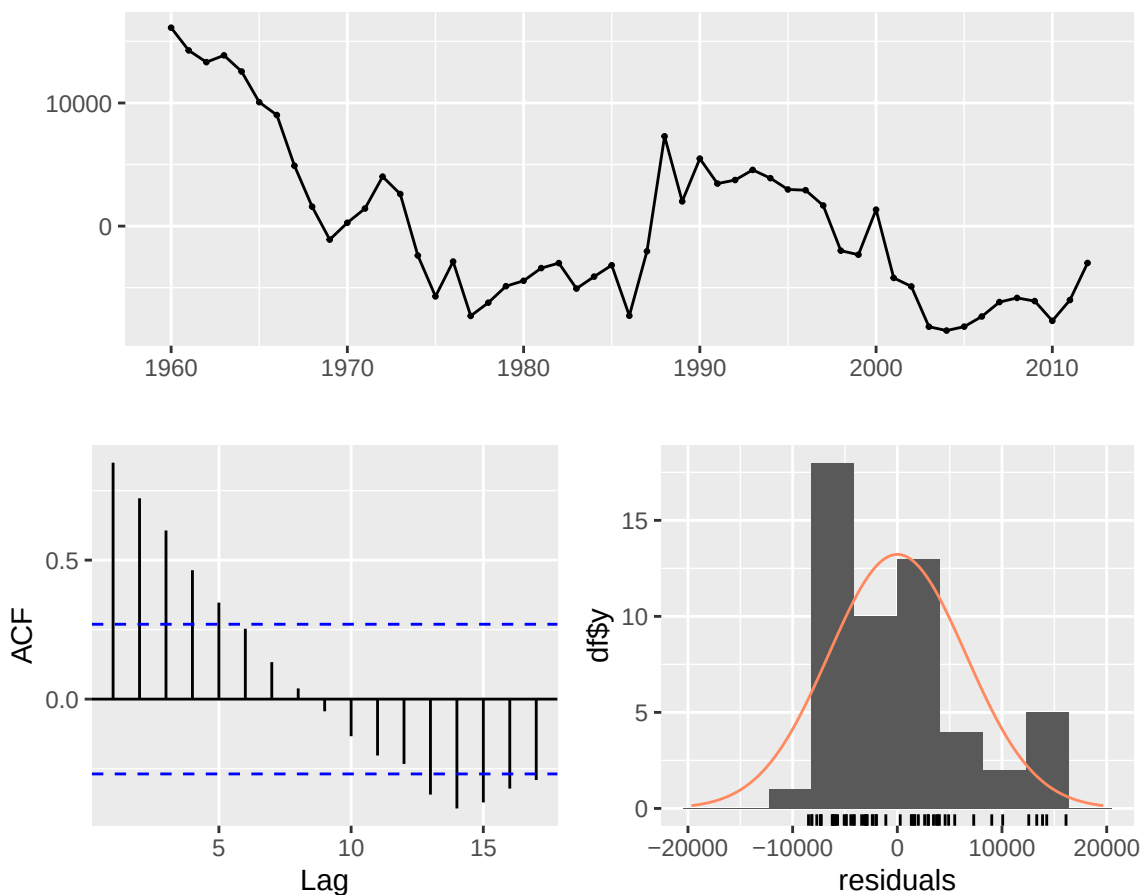


```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,0,0) with non-zero mean
## Q* = 201.64, df = 10, p-value < 2.2e-16
##
## Model df: 0.   Total lags used: 10
```

TLB - Candidate Model

```
## Series: tlb_train_ts
## ARIMA(0,0,0) with non-zero mean
##
## Coefficients:
##          mean
##      45659.3208
## s.e.      891.8053
##
## sigma^2 = 42962330: log likelihood = -540.46
## AIC=1084.92  AICc=1085.16  BIC=1088.86
##
## Training set error measures:
##              ME      RMSE      MAE      MPE      MAPE      MASE
## Training set -6.590665e-12 6492.436 5410.766 -1.887957 11.73191 2.585078
##              ACF1
## Training set 0.850366
```

Residuals from ARIMA(0,0,0) with non-zero mean



```
##
## Ljung-Box test
##
```

```
## data:  Residuals from ARIMA(0,0,0) with non-zero mean
## Q* = 118.42, df = 10, p-value < 2.2e-16
##
## Model df: 0.    Total lags used: 10
```
